Amirhossein Asadi

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RESEARCH INTERESTS

Learning-enabled control of nonlinear robotic systems with formal safety guarantees. I combine **reinforcement learning** and **model predictive control** with **control barrier functions** and **reachability analysis**, with applications to legged robots and human-interactive robots such as lower-limb exoskeletons.

EDUCATION

North Carolina State University 2024 – Present
M.S. in Mechanical Engineering; Minor in Electrical and Computer Engineering
Raleigh, NC · GPA 3.93/4.00 · ECE minor GPA 4.00/4.00

Iran University of Science and Technology 2019 – 2024
B.Sc. in Mechanical Engineering
Tehran, Iran · GPA 3.90/4.00

PUBLICATIONS

1. **A. H. Asadi** and N. Sharma, "Reduced-Order Backstepping Control of Nonlinear Mechanical Systems with Input Delay via Galerkin Projection," *IEEE Conference on Decision and Control (CDC)*, 2026.
2. **A. H. Asadi**, K. Lambeth, and N. Sharma, "SafeShadow: A Neural Barrier Safety Framework for Balance Recovery in Exoskeleton Systems," *IEEE Robotics and Automation Letters (RA-L)*, 2026. UNDER REVIEW
3. **A. H. Asadi** and N. Sharma, "Safety-Aware Dimension Selection for Reduced-Order Model Predictive Control," *IEEE Control Systems Letters (L-CSS)* with the *American Control Conference (ACC)* 2027 option. UNDER REVIEW
4. **A. H. Asadi**, N. Nikseresht, and A. H. Korayem, "Adaptive MIMO PID Control of a Wheel-Leg Manipulator by Considering the Slippage Based on Deep Reinforcement Learning," *11th RSI International Conference on Robotics and Mechatronics (ICRoM)*, IEEE, 2023, pp. 437–442. DOI
5. M. H. Korayem, N. Nikseresht, and **A. H. Asadi**, "Vision-Based Dynamic Modeling of Wheeled-Legged Robot Considering Slippage Using Gibbs–Appell Formulation," *Robotica*, 2024. DOI
6. **A. H. Asadi** and N. Sharma, "Reactive Stepping Under Delay: Hybrid Reach–Avoid Recovery for Bipedal Locomotion." IN PREPARATION
7. **A. H. Asadi** and N. Sharma, "Contraction–Barrier Recovery Funnels for Safe Reactive Stepping in Bipedal Locomotion." IN PREPARATION

RESEARCH EXPERIENCE

Graduate Student Researcher 2024 – Present
Neuromuscular Control and Robotics Laboratory, North Carolina State University

- Develop safety-critical and learning-based controllers for lower-limb exoskeletons, focused on balance recovery after external perturbations.
- Design formal safety certificates that guarantee safe recovery while integrating learned components into the control framework.

Undergraduate Research Assistant 20XX – 20XX
Intelligent Robotics Laboratory, Iran University of Science and Technology

- Worked at the intersection of control theory and reinforcement learning, with applications to wheeled-legged and flexible manipulators.
- Built dynamic models and controllers for complex robots, including learning-enabled control of a wheeled-legged manipulator [4, 5].

SELECTED RESEARCH PROJECTS

MODEL-BASED CONTROL

Reduced-Order Control of Nonlinear Systems with Input Delay [1]

- Modeled actuator delay as a transport PDE and derived a Galerkin reduced-order representation for controller synthesis.
- Designed a backstepping controller for nonlinear mechanical systems, proved closed-loop stability with Lyapunov analysis, and developed a command-filtered implementation.

Safety-Aware Reduced-Order Model Predictive Control [3]

- Built an adaptive-order MPC framework combining dynamic active subspaces, reduced-order prediction, and discrete-time control barrier functions.
- Designed a safety-driven rank-selection rule that raises the prediction dimension only when needed to preserve finite-horizon safety and recursive feasibility.

Reachability-Based Reactive Stepping for Bipedal Robots [6]

- Formulated perturbation recovery as a delay-aware reach–avoid game and decomposed it into lower-dimensional pendular and swing-leg subproblems.
- Derived certified walking terminal sets and feasibility conditions for reactive steps under delayed control authority.

Hybrid Contraction–Barrier Recovery for Bipedal Locomotion [7]

- Developed a framework that turns transverse contraction geometry and physical safety clearance into phase-dependent recovery funnels around a walking gait.
- Used saltation-based impact sensitivity to certify hybrid recovery and trigger reactive steps, choosing footholds by post-impact recovery margin and building recovery sets recursively.

LEARNING-BASED CONTROL

Neural Safety Certificates for Exoskeleton Balance Recovery [2]

- Trained a neural high-order control barrier function to learn posture- and actuation-dependent balance-feasible sets for lower-limb exoskeletons.
- Embedded the learned certificate in a CLF–CBF quadratic program for real-time balance preservation and constrained allocation between motor actuation and functional electrical stimulation (FES).

Deep Reinforcement Learning for Wheel-Leg Manipulator Control [4]

- Used deep deterministic policy gradient (DDPG) to adapt the gains of a MIMO PID controller for a wheeled-legged manipulator.
- Incorporated wheel slippage into the inverse-kinematic model to improve robustness to locomotion uncertainty.

Learning-Based Control of a Flexible Revolute–Prismatic Manipulator

- Developed dynamic models and MATLAB simulations targeting end-effector regulation and vibration suppression.
- Compared PID, fuzzy-PID, Q-learning, and multi-agent reinforcement learning, including RL-based tuning of PID gains and fuzzy rules.

Reinforcement Learning for Robotic Control

- Implemented SAC and DDPG agents in simulation for 7-DoF manipulator ball balancing, quadruped locomotion, and adaptive cruise control.

ROBOTIC SYSTEMS AND HARDWARE

Lower-Limb Exoskeleton Balance-Recovery Platform

- Developed an exoskeleton framework for perturbation recovery that integrates rigid-body dynamics, contact constraints, actuator limits, and center-of-mass regulation.
- Combined robotic actuation with FES and evaluated safety-critical controllers in simulation and in experimental balance-recovery trials.

TECHNICAL SKILLS

Control and theory	Nonlinear control, MPC, control barrier functions, Lyapunov analysis, reachability analysis, optimization
Learning	Deep RL (DDPG, SAC), Q-learning, multi-agent RL, neural safety certificates
Programming	MATLAB, Python, C/C++, Julia
Robotics and simulation	ROS 2, MuJoCo, Gazebo, Simulink, OpenCV
Tools	Git, LaTeX, Linux, Docker
Hardware	Sensors, embedded systems (Raspberry Pi, Arduino), CAN bus, real-time control

HONORS AND AWARDS

Fellowship, Iran's National Elites Foundation (INEF) 2022–2023

1st Place, SuperTeam Challenge, RoboCup IranOpen, Tehran 2023

1st Place, Autonomous Cars League, FIRA National Competition, Iran 2023

Top 0.06% nationwide, Iranian University Entrance Examination

Exceptional Talents admission: admitted to M.S. study without the entrance examination

TEACHING

North Carolina State University

MAE 306: Thermal-Fluid Sciences Lab — Fall 2025, Spring 2026

MAE 253: Experimental Aerodynamics Lab — Spring 2025

MAE 371: Aerospace Vehicle Structures Lab — Fall 2024

Iran University of Science and Technology

Linear Control Systems, Department of Railway Engineering — Winter 2023

Dynamics, Department of Mechanical Engineering — Fall 2022

Mechanical Vibrations, Department of Mechanical Engineering — Winter 2022

Instructor, MATLAB/Simulink and SolidWorks courses and workshops

SERVICE AND LEADERSHIP

Reviewer, IEEE Robotics and Automation Letters (RA-L); American Control Conference (ACC)

President, Robotics and Mechatronics Student Association, Iran University of Science and Technology

Team Leader and Programmer, RoboCup Autonomous Cars League

- Led the team in designing and building an autonomous mobile robot for competition.

LANGUAGES

English	TOEFL iBT 102/120 (Reading 29, Listening 28, Speaking 23, Writing 22)
Persian (Farsi)	Native